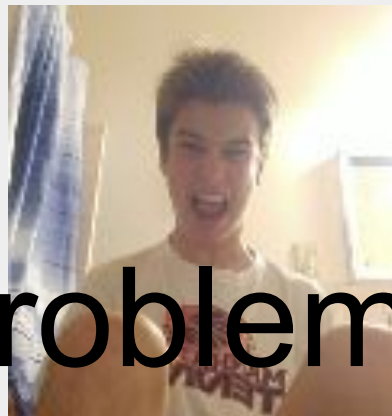




Birthday Problem



HAPPY BIRTHDAY SYOMA



The Problem!

How many people must be in a room to make it likely that 2 people have the same birthday?



Let's Solve it!

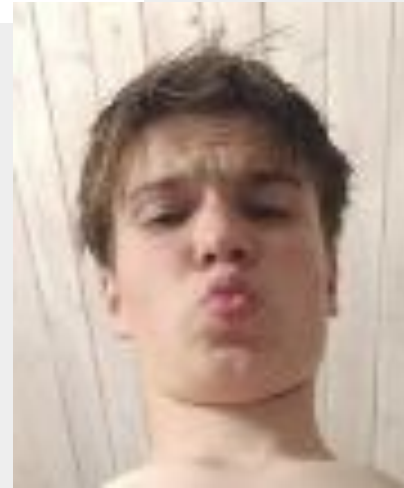
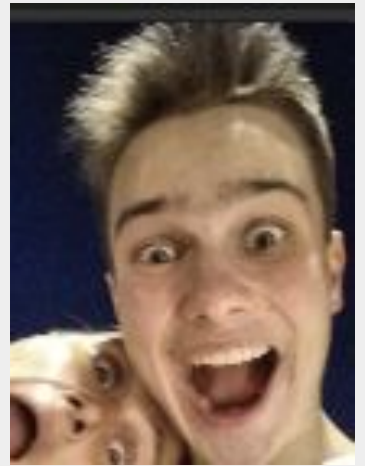
How should we approach this problem?

The probability $P(n)$ that at least two people in a group of n share a birthday is:

$$P(n) = 1 - \underbrace{\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{365 - n + 1}{365}}_{\text{Probability all birthdays are unique}}$$

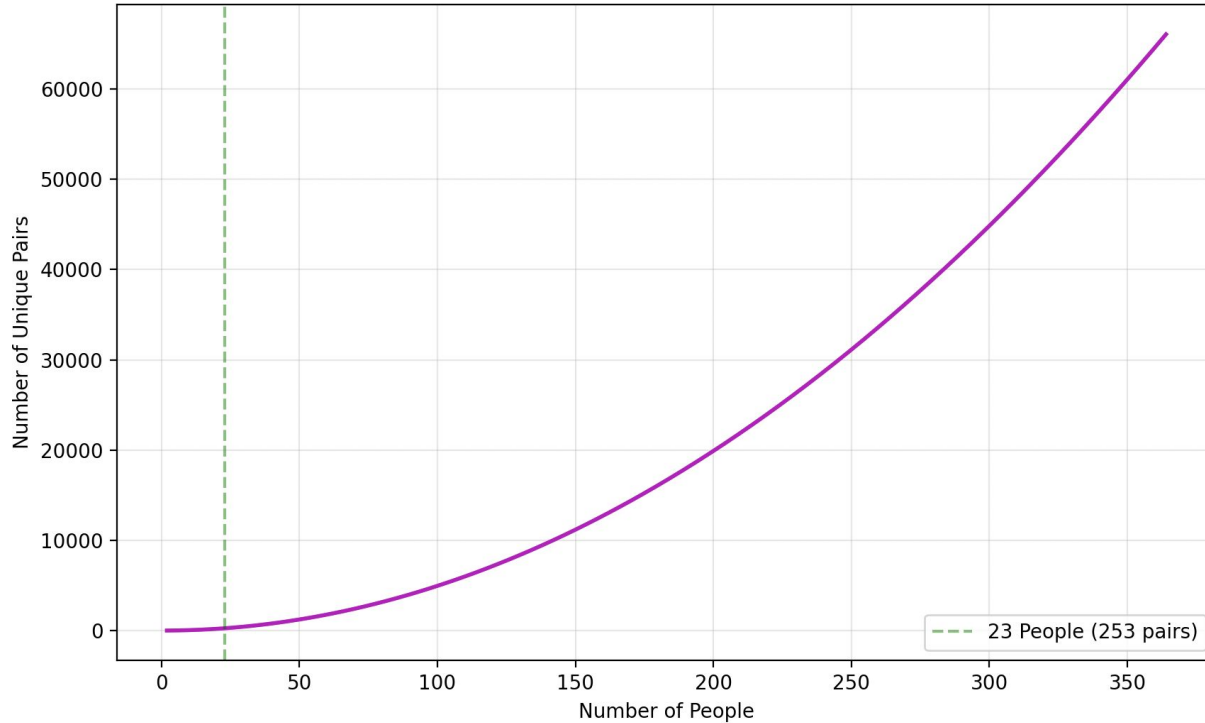
To find the smallest n where $P(n) > 0.5$, solve:

$$\prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right) < 0.5$$

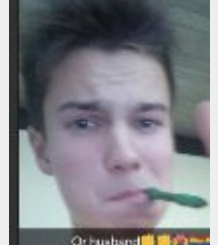
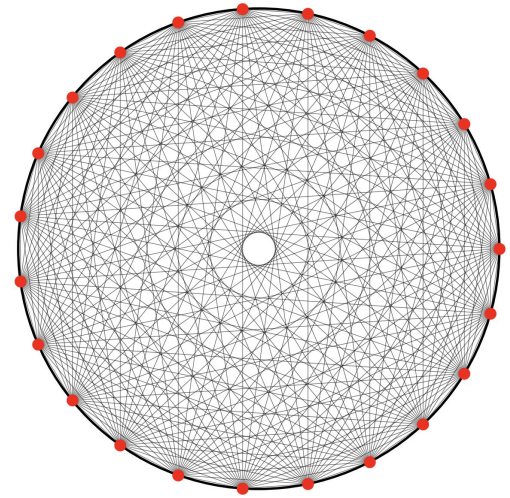


Why it like this?

Quadratic Growth of Possible Birthday Matches



23 People: All Possible Connections (253 Pairs)



Similar Problems!

How many people are needed for a $>99\%$ chance of a shared birthday?



What's the probability that in a group of 23 people, exactly two people share a birthday?

